

## Project Planning

|               |                                             |
|---------------|---------------------------------------------|
| Date          | 12 July 2026                                |
| Team ID       |                                             |
| Project Name  | Emotion Detection & Learning Support Engine |
| Maximum Marks | 5 Marks                                     |

### Product Backlog, Sprint Schedule, and Estimation

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                         | Story Points | Priority | Team Members |
|----------|-------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------|--------------|----------|--------------|
| Sprint-1 | Emotion Detection             | USN-1             | As a student, I can submit free text describing my problem so the system can analyse it.                                  | 2            | High     | Individual   |
| Sprint-1 | Emotion Detection             | USN-2             | As a system, I can classify submitted text into one of five emotions using a BiLSTM model for a fast baseline prediction. | 3            | High     | Individual   |
| Sprint-1 | Emotion Detection             | USN-3             | As a system, I can classify submitted text using a BERT model so predictions can be cross-checked against BiLSTM.         | 3            | High     | Individual   |
| Sprint-1 | AI Response Generation        | USN-4             | As a student, I can receive personalised guidance text based on my detected emotion.                                      | 3            | High     | Individual   |
| Sprint-1 | AI Response Generation        | USN-5             | As a system, I can fall back to an offline mock response if the Gemini API is unavailable.                                | 2            | Medium   | Individual   |
| Sprint-2 | Logging & Analytics           | USN-6             | As a system, I can log every interaction (input, emotion, confidence, response) to CSV for later analysis.                | 2            | High     | Individual   |
| Sprint-2 | Logging & Analytics           | USN-7             | As a student, I can view a live analytics dashboard showing emotion distribution and model confidence.                    | 3            | Medium   | Individual   |

|          |                      |        |                                                                                                                                                   |   |        |            |
|----------|----------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|------------|
| Sprint-2 | Frontend             | USN-8  | As a student, I can use a simple Streamlit interface to type my problem and click "Get Guidance".                                                 | 2 | High   | Individual |
| Sprint-2 | Frontend             | USN-9  | As a student, I can see my detected emotion and confidence percentage displayed clearly.                                                          | 1 | Medium | Individual |
| Sprint-3 | Deployment & Testing | USN-10 | As a team, we can deploy the app locally via a virtual environment and verify all five emotion categories produce distinct, appropriate guidance. | 2 | Medium | Individual |